

JEE MAIN REVISION MASTER: THREE DIMENSIONAL GEOMETRY 2026 3 QUESTIONS 5% WEIGHTAGE

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 41 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Direction Cosines & Equations of Lines in 3D	23 PYQs	3D Lines
Very High	Planes & Line-Plane Intersections in 3D	19 PYQs	3D Planes

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Direction Cosines & Equations of Lines in 3D	$l^2 + m^2 + n^2 = 1$, $\frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - z_1}{c}$, $d_{\text{skew}} = \frac{ \vec{a}_2 \cdot (\vec{b}_1 \times \vec{b}_2) }{ \vec{b}_1 \times \vec{b}_2 }$
Planes & Line-Plane Intersections in 3D	$a(x - x_1) + b(y - y_1) + c(z - z_1) = 0$, $d = \frac{ ax_1 + by_1 + cz_1 + d }{\sqrt{a^2 + b^2 + c^2}}$, $\sin\theta = \frac{ \vec{b} \cdot \vec{n} }{ \vec{b} \vec{n} }$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Direction Cosines & Equations of Lines in 3D (Very High Yield — 23 PYQs)

Core Principle: Direction cosines (l, m, n), line passing through two points, and shortest distance formula between skew lines.

■ **Caution / JEE Pitfall:** Direction ratios (a, b, c) can be any proportional numbers, but direction cosines (l, m, n) MUST satisfy $l^2 + m^2 + n^2 = 1$.

■ **Speed Trick:** Two lines in 3D intersect (are coplanar) if and only if the shortest distance between them is ZERO: $(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2) = 0$.

Representative Source PYQs:

- [2026 Shift] Q#2 (MCQ): "The measure of the angle between the lines $x=k+1, y=2k-1, z=2k+3, k \in \mathbb{R}$ and $(x-1)/2=(y-2)/1=(z-3)/1$ is..." [Ans: B]
- [2026 Shift] Q#3 (MCQ): "The angle between the lines whose direction ratios are a,b,c and b-c,c-a,a-b is..." [Ans: N/A]
- [2025 Shift] Q#2 (MCQ): "The distance of the point P(-3,4,5) from yz plane is..." [Ans: N/A]

3.2 Planes & Line-Plane Intersections in 3D (Very High Yield — 19 PYQs)

Core Principle: Normal vector to a plane, perpendicular distance from point to plane, and angle between a line and plane (uses $\sin\theta$).

■ **Caution / JEE Pitfall:** Angle between a LINE (vector $\vec{b}$) and a PLANE (normal $\vec{n}$) is computed with SINE ($\sin\theta = \frac{|\vec{b} \cdot \vec{n}|}{|\vec{b}||\vec{n}|}$), NOT cosine!

■ **Speed Trick:** A line is parallel to a plane if and only if it is perpendicular to the plane's normal: $\vec{b} \cdot \vec{n} = 0$.

Representative Source PYQs:

- **[2026 Shift] Q#1 (MCQ):** "The three points $A(2,4,3), B(4,a,9)$ and $C(10,-1,7)$ form a right-angled triangle with $\angle B = 90^\circ$, then the value of "a" is..." [Ans: B]
- **[2025 Shift] Q#1 (MCQ):** "The equation of the line through the point $(0,1,2)$ and perpendicular to the line $(x-1)/2 = (y+1)/3 = (z-1)/-2$ is..." [Ans: N/A]
- **[2024 Shift] Q#1 (MCQ):** "The sine of the angle between the straight line $(x-2)/3 = (y-3)/4 = (z-4)/-5$ and the plane $2x-2y+z=5$ is..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

The three points $A(2,4,3), B(4,a,9)$ and $C(10,-1,7)$ form a right-angled triangle with $\angle B = 90^\circ$, then the value of "a" is

- (A) 1 or 4
- (B) -1 or 4 [Correct]
- (C) 1 or -4
- (D) -1 or -4

Correct Answer: B

Solution Analysis: Since the triangle is right-angled at B, the two sides meeting at B must be perpendicular. So, vectors $\vec{BA}$ and $\vec{BC}$ must satisfy $\vec{BA} \cdot \vec{BC} = 0$. Now, $A(2,4,3), B(4,a,9), C(10,-1,7)$. So, $\vec{BA} = A - B = (2-4, 4-a, 3-9) = (-2, 4-a, -6)$. Also, $\vec{BC} = C - B = (10-4, -1-a, 7-9) = (6, -1-a, -2)$. Since they are perpendicular, $(-2)(6) + (4-a)(-1-a) + (-6)(-2) = 0$. $-12 + (4-a)(-1-a) + 12 = 0$. $(4-a)(-1-a) = 0$. So, $4-a=0$ or $-1-a=0$. Hence, $a=4$ or $a=-1$. So the correct option is Option B-1 or 4.

Benchmark Problem 2: KCET 2026 (Q#2)

The measure of the angle between the lines $x=k+1, y=2k-1, z=2k+3, k \in \mathbb{R}$ and $(x-1)/2 = (y-2)/1 = (z-3)/1$ is

- (A) $\cos^{-1}((2/3))$
- (B) $\cos^{-1}(\sqrt{(2/3)})$ [Correct]
- (C) $\cos^{-1}(\sqrt{(3/2)})$
- (D) $\cos^{-1}(3/2)$

Correct Answer: B

Solution Analysis: For the first line, $x=k+1, y=2k-1, z=2k+3$. The direction ratios are the coefficients of k . So, direction ratios of the first line are 1, 2, 2. For the second line, $(x-1)/2 = (y-2)/1 = (z-3)/1$. The direction ratios are 2, 1, 1. The angle θ between two lines with direction ratios (a_1, b_1, c_1) and (a_2, b_2, c_2) is given by $\cos\theta = \frac{|a_1a_2 + b_1b_2 + c_1c_2|}{\sqrt{a_1^2 + b_1^2 + c_1^2} \sqrt{a_2^2 + b_2^2 + c_2^2}}$. Here, $\cos\theta = \frac{|1 \cdot 2 + 2 \cdot 1 + 2 \cdot 1|}{\sqrt{1^2 + 2^2 + 2^2} \sqrt{2^2 + 1^2 + 1^2}} = \frac{5}{\sqrt{9} \sqrt{6}} = \frac{5}{3\sqrt{6}}$. So, $\theta = \cos^{-1}(\sqrt{2/3})$.